

Stuti Shah

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EDUCATION

Virginia Tech

Master of Engineering in Computer Science

Blacksburg, VA

Aug 2025 – May 2026

Virginia Tech

Bachelor of Science in Computer Science, Minor in Economics

Blacksburg, VA

Aug 2022 – May 2025

EXPERIENCE

Software Engineering / Data Science Intern

May 2025 – Aug 2025

Volvo Group

Greensboro, NC

- Built backend and analytics services using Python, Flask, PostgreSQL, Docker, Kubernetes, and Argo CD, improving reliability, deployment reproducibility, and maintainability across engineering workflows.
- Automated DMIS output file generation using Streamlit, reducing processing time from 4 hours to under 6 minutes and delivering \$176,980 in annual operational savings across distributed engineering teams.
- Designed an end-to-end ML analytics platform linking complete vehicle modeling signals with millions of vehicle sensor records for scalable predictive performance analysis and engineering decision support.
- Developed REST APIs to connect high-volume vehicle test data with analytics pipelines, improving data traceability, query access, and workflow execution speed.
- Built and validated Random Forest models to predict truck component damage and acceleration, supporting data-driven test readiness decisions across vehicle platforms and configurations.
- Integrated predictive models into analytics workflows, including component performance prediction and GHG emissions estimation.
- Built a GHG emissions prediction pipeline in Power BI by estimating missing CdA values and enabling scenario-based emissions classification for sustainability and regulatory reporting.
- Deployed services through GitOps-based infrastructure using Kubernetes and Argo CD, ensuring reproducible environments across development, staging, and production-aligned workflows.

AI Intern

Jun 2024 – Aug 2024

AquaOrange

Bangkok, Thailand

- Automated test workflows for 10+ Zoho CRM customizations using BrowserStack and testRigor, improving regression coverage across frontend and backend features.
- Built validation pipelines for customized CRM workflows, reducing manual QA effort and improving defect detection during release cycles.

Data Science Intern / Solutions Associate

Jun 2023 – Jul 2024

HERE! Wireless USA

Dallas, TX

- Executed 50+ alpha testing workflows across Android and iOS applications, improving parental control feature stability and reducing post-release issues.
- Conducted root-cause analysis for mobile app issues, helping improve feature reliability and user experience across product iterations.
- Collaborated with product managers and engineers to translate customer feedback into actionable product and engineering requirements.
- Developed four pricing models to support leadership in minimizing revenue leakage, improving market positioning, and optimizing product strategy.

CS Undergraduate Teaching Assistant

Aug 2024 – May 2025

Virginia Tech

Blacksburg, VA

- Mentored 200+ students in computer systems topics including processor architecture, assembly programming, memory hierarchy, caching, and performance optimization.
- Debugged low-level programming assignments involving register usage, memory access patterns, instruction execution, stack behavior, and control flow.
- Guided students through system-level reasoning concepts including cache behavior, instruction-level execution, processor pipelines, and execution-time analysis.

- Supported students through office hours, debugging sessions, and assignment guidance focused on low-level software and hardware interactions.

Director of Finance

Jan 2025 – Jan 2026

CS Careers @ VT

Blacksburg, VA

- Managed finance-related operations for a student organization focused on supporting computer science career development and professional opportunities.
- Coordinated budgeting and organizational planning to support events, career initiatives, and student engagement activities.
- Collaborated with student leaders to support programming that connects students with career resources, industry opportunities, and technical growth.

Division I Badminton Athlete

Aug 2022 – May 2026

Virginia Tech

Blacksburg, VA

- Singles runner-up in Division I collegiate competition, demonstrating consistency and competitive execution under pressure.
- Contributed to team championship wins and a third-place Division I collegiate team finish across tournament play.
- Helped the team secure qualification for nationals through strong performances in high-stakes matches.
- Developed composure, resilience, and decision-making in pressure-heavy tournament environments.

PROJECTS

Multimodal Uncertainty-Aware Stock Return Forecasting After Earnings Calls

- Built a multimodal forecasting pipeline for next-trading-day post-earnings log returns using S&P 500 earnings call transcripts, structured financial indicators, and Reddit-based sentiment features.
- Encoded long earnings call transcripts with ProsusAI FinBERT using 256-512 token chunks, cached frozen transcript embeddings, and projected text representations into a shared multimodal embedding space.
- Engineered structured inputs from standardized earnings surprise, price momentum, and implied volatility, and aggregated Reddit sentiment with TextBlob polarity and log message-volume features.
- Implemented a PyTorch cross-modal fusion model using multi-head attention across transcript, financial, and sentiment modality tokens, with quantile prediction heads trained using pinball loss.
- Applied conformalized quantile regression with global, event-conditioned, and modality-disagreement-normalized calibration to generate 80%, 90%, and 95% prediction intervals.
- Evaluated raw multimodal, single-modality ablations, conformal variants, and same-ticker historical baselines using empirical coverage, interval width, calibration error, MAE, RMSE, and directional accuracy.

YOLO11 Shelf Detection for Planogram Compliance

- Built an end-to-end shelf-auditing system using YOLO11, OpenCV, and Qwen2-VL to detect products, identify empty shelf spaces, extract SKU-level details, and assess planogram compliance.
- Deployed the application on AWS EC2, supporting image-based audit processing for retail inventory and shelf-compliance use cases.
- Designed planogram comparison logic to classify shelf items as correctly stocked, missing, misplaced, or unverified for retail restocking workflows.
- Developed Flask backend APIs to process shelf images, run detection and visual analysis pipelines, and return structured audit results to a React dashboard.
- Curated public shelf-image datasets, trained detection models, and analyzed failure cases involving occlusion, partial visibility, visually similar products, and inconsistent planogram data.

CS Source JobHub

- Developed backend services and REST APIs using Flask, SQLAlchemy, and MySQL to support employer registration, contact management, membership lifecycle tracking, and role-based workflows.
- Designed relational database models for employers, contacts, contact types, billing details, and membership records with validation logic to improve data consistency and integrity.
- Implemented backend workflows for form submission, data persistence, authentication-related logic, and structured employer/contact management.

Location-Aware Task Management App

- Built an MVVM-based Android application using Java, Room, LiveData, and ViewModel, enabling offline-first CRUD workflows for creating, updating, deleting, and managing location-aware tasks.

- Implemented local database persistence and geolocation-based task features to support reliable task management without continuous network access.
- Designed responsive user workflows for task creation, task updates, task deletion, and location-aware task organization.

HybridDiff AR: Efficient Image Generation Pipeline

- Developed a performance-focused image generation pipeline combining a 33M-parameter DDPM with transformer-based refinement for efficient CelebA 64x64 image generation.
- Optimized the inference workflow to reduce generation latency while preserving image quality, achieving up to 250x faster inference in experimental evaluation.
- Evaluated model performance with attention to inference speed, image quality, and practical deployment tradeoffs.

TECHNICAL SKILLS

Programming: Python, Java, C, SQL

AI & ML: AI Agents, Prompt Engineering, Feature Engineering, Model Evaluation

AI/ML Libraries: PyTorch, scikit-learn, TensorFlow

Backend & Systems: Flask, REST APIs, Linux, System Design Fundamentals, Memory Management

Databases & Data Engineering: PostgreSQL, MySQL, SQLAlchemy, Spark/PySpark, ETL Pipelines, Query Optimization

Cloud: AWS Bedrock, SageMaker, AWS Redshift, AWS S3/Blob Storage, AWS Lambda, AWS Functions, Azure Databricks, Azure SQL, AWS EC2

DevOps: Docker, Kubernetes, Argo CD, GitOps, CI/CD, Git

BI & Analytics: Dashboard Storytelling, KPI Frameworks, Streamlit Analytics, Power BI, Tableau, Data Visualization

Software Engineering: OOP, Data Structures & Algorithms, Unit Testing, Debugging, Logging & Monitoring